



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECR 1213

Network Communication

Task 2 – Initial Design

Preliminary Analysis

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1.0 Questions

Before developing the detailed network design, it is important to identify the critical requirements for infrastructure, connectivity, security, and specialized facilities. The following questions guide this analysis and ensure that the proposed solution is both feasible and aligned with the faculty's long-term goals.

1. Is it important to have a server room within the building?

A dedicated server room is essential because it provides a secure and controlled environment for critical network equipment. This room houses servers, core switches, firewalls, storage systems, and backup devices that support the building's ICT operations. It must include environmental monitoring systems to maintain proper temperature and humidity, ensuring reliable equipment performance. Security measures such as restricted access, surveillance, and network protection systems prevent unauthorized access and cyber threats. By centralizing these functions, the building ensures better data protection, easier maintenance, and stable network performance for all labs and classrooms.

2. What are the required specifications for high-speed internet connections in the labs, classrooms, and video conferencing rooms?

To support smooth computing activities across all labs and classrooms, each space must be equipped with stable high-speed internet connectivity. A gigabit backbone connection should link the server room and distribution switches, providing sufficient capacity for large volumes of traffic. For end-user connections, a minimum of 100 Mbps per workstation is recommended in general labs to accommodate tasks such as programming, simulations, and online collaboration. The video conferencing room requires at least 50–100 Mbps of dedicated bandwidth to support HD streaming and hybrid meetings without lag. To prevent congestion during peak usage, a bandwidth management system should be implemented, allowing dynamic allocation of resources and prioritization of essential traffic. This setup ensures that all rooms, including labs, the hybrid classroom, and the video conferencing room, experience fast and reliable connections during teaching and practical sessions.

3. How can the wireless network coverage be optimized to support all areas of the faculty?

Optimizing wireless coverage requires a structured approach that considers the building layout, wall materials, and user density. A comprehensive site survey should be conducted to identify signal strengths, overlapping coverage, and potential dead zones. Predictive mapping tools can be used to visualize signal distribution and highlight weak spots caused by structural barriers. Access points must then be strategically placed in corridors, student lounges, labs, and open areas to eliminate weak spots, while dual-band WiFi (2.4 GHz and 5 GHz) balances speed and coverage.

After deployment, a physical validation survey should be carried out, where technicians walk through the facility with laptops and surveying software to confirm coverage. This engineered approach guarantees seamless connectivity across both floors, supporting high-density usage and ensuring students and staff remain connected throughout the building.

4. What type of network cable is the most appropriate for connecting the computers, and what are the reasons for selecting it?

The most appropriate network cable for connecting computers in this project is the Cat6 Ethernet cable. Cat6 supports speeds up to 1 Gbps over distances of 100 meters and up to 10 Gbps at shorter distances, which is sufficient for most educational applications. It also provides better shielding against electromagnetic interference, a common issue in school environments with numerous electronic devices. Cat6 cables are cost-effective, widely available, and compatible with existing standards, making them a practical choice for large-scale deployment. Additionally, they offer a degree of future-proofing, as they can support higher speeds if the institution upgrades its backbone infrastructure later. This balance of performance, reliability, and affordability makes Cat6 the most suitable option for the faculty's labs and classrooms.

5. What security measures are essential to protect the faculty's network infrastructure?

Security is a critical requirement for the faculty's network, given the sensitivity of academic data and the need to protect against breaches. Essential measures include deploying next-generation firewalls with intrusion prevention systems to block malicious traffic, implementing VLAN segmentation to separate student, staff, and administrative traffic, and enforcing secure authentication protocols such as WPA3 and multi-factor login for critical systems. End-to-end encryption should be applied to sensitive communications, while real-time monitoring tools provide alerts for unusual traffic patterns. Regular updates and patching of routers, switches, and servers are also necessary to prevent exploitation of vulnerabilities. Together, these measures ensure the faculty's network remains resilient, secure, and compliant while supporting high-performance teaching and research activities.

6. How can redundancy and backup systems be integrated into critical areas such as the video conferencing room and all labs to ensure data integrity and availability?

Redundancy and backup systems are vital to ensure that critical rooms, including the video conferencing room and labs, can continue operating even if failures occur. One approach is to provide each room with multiple network connections so that if the main link is interrupted, the system automatically switches to a backup link. Important rooms such as the Cisco Network Lab and hybrid classroom can also use additional

switches or access points, ensuring there is always a backup device ready if one becomes faulty. For data protection, the building should perform regular scheduled backups of server data, system configurations, user profiles, and lab materials. These backups should be stored both onsite and offsite to protect against hardware failure, theft, or disasters. Implementing RAID storage inside the server room further enhances resilience by distributing data across multiple drives. With these measures in place, the building can recover quickly from failures and continue operating without major interruptions.

7. What specific essential hardware will be used in Cisco Network Lab?

The specific essential hardware that will be in a Cisco Network Laboratory will be routers, switches, wireless access point and security appliances.

- I. Routers To learn routing protocols (OSPF, EIGRP, BGP), WAN technologies, VPNs (IPsec, SSL), and Network Address Translation (NAT). The recommended router is Cisco 1100 Series ISR (e.g., C1111-4P) or 900 Series ISR.
- II. Layer 3 switches, which is core of the lab, are also essential to learn VLANs, Trunking, STP, EtherChannel, Switch Stacking, and basic Layer 3 routing. The recommended switch is Cisco 2960-S or 3560-G.
- III. Wireless access points for students to learn WLAN design. APs allow teaching of channel planning, roaming, and multiple SSIDs, and recommend using Cisco Aironet 1800/2800 Series (WiFi 5/6).
- IV. Security appliances are also a must for integrated security. Next-Generation Firewall (NGFW) with intrusion prevention (IPS). Critical for teaching modern threat defense and security policies.

8. How many people on average will be at the lounge and pantry using the WiFi during peak hours?

In accordance with the normal number of students and staff in an area like the lounge expects to have an average of fifteen to twenty users busy at any one time. This estimation makes it easier to quantify the density needed for Wi-Fi access points so as to get the right density to support the coverage and speed needed.

9. What tools can be used for real-time network monitoring and performance tracking?

Real-time network monitoring can be achieved using tools such as ManageEngine OpManager, Obkio, and PRTG Network Monitor, which provide continuous visibility into bandwidth usage, device health, latency, packet loss, and overall network performance. OpManager enables real-time tracking of switches, routers, servers, and wireless devices through SNMP polling and threshold-based alerts, while Obkio uses cloud-based monitoring agents to measure jitter, latency, and packet loss between

network segments, providing detailed insight into end-to-end performance. Meanwhile, PRTG offers customizable sensors for traffic flow, CPU load, and interface utilization, making it suitable for monitoring large academic networks with multiple labs. These tools allow administrators to quickly detect issues, analyze trends, and ensure that the network remains stable and efficient under varying workloads.

10. What are the essential network and hardware requirements for a high-quality video conferencing room?

A high-quality video conferencing room requires a combination of reliable network connectivity and professional-grade hardware to ensure smooth communication. At minimum, the room should have a stable high-speed wired connection with at least 4–10 Mbps per active video stream to support HD or 4K conferencing, along with low latency and minimal packet loss to avoid disruptions. Essential hardware includes a high-resolution webcam or PTZ camera, ceiling or boundary microphones, echo-cancelling speakers, and a dedicated conference codec or software platform capable of handling high-definition video. Additionally, proper lighting, acoustic treatment, and a backup connection help maintain consistent quality during sessions. These components collectively ensure clear audio, sharp video, and a professional conferencing experience suitable for academic or administrative use.

2.0 Feasibility Assessment

After finding out the answers of the questions regarding the requirements to develop a network plan, we can now assess the feasibility of this new building project for the Faculty of Computing. The feasibility assessment involves evaluating the technical, economic and operational aspects of the project to determine whether this is viable.

2.1 Technical Feasibility

2.1.1 Project Description & Scope

The project involves the construction of a two-storey building (42m x 26m) and the implementation of its core technological infrastructure, including:

- 4 specialized computing labs (30 workstations each).
- 1 high-fidelity video conferencing room.
- 1 hybrid classroom.
- 1 central server room (~40 sq m).
- Campus-wide Wi-Fi coverage within the building.
- Supporting spaces (offices, student lounge).

2.1.2 Technology Availability & Suitability

- Network Hardware: Core technologies (Cisco networking gear, Wi-Fi 6 APs) are standard, widely available, and proven in educational settings. The proposed server room design (16 racks) is scalable.
- Lab Equipment: Specialized lab requirements (e.g., Cisco networking pods, high-performance computing workstations) can be sourced from established vendors. The mix of physical and virtualized labs (using Cisco CML/Packet Tracer) is a cost-effective and scalable approach.
- Hybrid Technology: Integrated Audio-Visual systems for hybrid learning and video conferencing (e.g., Video Conferencing Room, Hybrid Room) are reliable and commercially available.

2.1.3 Technical Feasibility Conclusion

The project is technically feasible. All proposed technologies are mature, available, capable of supporting high-performance to the core backbone and can be integrated successfully with careful planning and expert implementation.

2.2 Economic Feasibility

2.2.1 Project Scope Adjustment based on Budget Constraint

The total project budget is fixed at MYR 2M. This budget is not solely for IT infrastructure and must also cover the fit-out, furnishing, and essential services for other critical areas, including offices, pantries, toilets, and the student lounge.

Therefore, the allocation for the IT and specialized lab infrastructure must be strategically constrained to likely no more than MYR 1.3M - 1.5M to leave adequate funding for the rest of the building. This necessitates a highly focused and cost-effective approach.

2.2.2 Estimated IT Infrastructure Costs

The IT budget is divided into two main parts:

- 1) Building-Wide Core Network
- 2) Specialized Lab & Room Fit-Out

Part 1: Building-Wide Core Network (The Shared Backbone)

This infrastructure is essential for the entire building to function.

Component	Estimated Cost (MYR)	Justification
Structured Cabling (Cat6 Ethernet)	180,000	Foundation for all connectivity. Cost to wire all rooms (offices, labs, classrooms) with data points, running back to a central server room.
Core Network Hardware (Switches, Firewall, Wi-Fi)	420,000	The network's brain. Includes a core switch, internet firewall, Wireless LAN Controller, and Wi-Fi 6 Access Points for seamless coverage everywhere.
Central Server Room (Racks, UPS, Cooling)	100,000	Hub for all systems. Basic server rack, a Unified Threat Management (UTM) appliance, and a sufficient UPS and cooling solution.
Sub-Total for Core Network	700,000	

Part 2: Specialized Lab & Room Fit-Out

These are the costs specific to the high-tech rooms.

Room	Estimated Cost (MYR)	Justification
Cisco Networking	450,000	Costs for specialized devices only. This

Lab		covers the routers (Cisco 1100/900 Series), Layer 3 switches (2960-S/3560-G), firewalls (Firepower 1000), and APs (Aironet 1800/2800) for hands-on pods. Excludes standard workstations.
Video Conferencing Room	150,000	High-quality AV integration. Covers PTZ camera, ceiling microphones, echo-cancelling speaker, video codec, acoustic treatment, and dedicated network switch for a stable, high-speed connection.
Embedded Systems Lab	100,000	Devices & peripherals for learning. Microcontrollers (e.g., ARM/FPGA boards), sensors, oscilloscopes, logic analyzers, and development kits.
Hybrid Classroom	100,000	Setup for teaching efficiency. Includes a tracking camera, superior audio system (microphones/speakers), interactive display, and control panel to seamlessly blend in-person and remote learning.
Sub-Total for Specialized Rooms	800,000	

Total IT infrastructure budget: MYR1.5M

Contingency (for IT, ~7%) : MYR0.1M

Grand total for IT: MYR1.6M

2.2.3 Financial Justification and 4IR Alignment

This allocation ensures the building is 4IR-ready without exceeding the overall budget.

Strategic Prioritization: The Cisco Lab receives the largest single-room investment as it is the primary, hands-on technical training facility.

Cost Control: By separating the core network cost, it is clear that the Cisco Lab's specific devices are a significant but justified investment. Using standardized Cat6 cabling and bulk purchasing for switches and APs keeps costs predictable.

Budget for Other Areas: This IT budget of MYR1.6M leaves a MYR0.4M buffer within the total MYR2.0M project budget to cover:

- Furniture, fixtures, and equipment for offices, student lounge, and pantry.
- Sanitary ware and fittings for toilets.
- Basic audiovisual equipment for standard classrooms.
- Construction and administrative costs.

2.2.4 Economic Feasibility Conclusion

The project is economically feasible within the MYR2.0M total budget. The proposed allocation of MYR1.6M for IT creates a modern technological foundation that directly supports the faculty's teaching mission, while the remaining MYR0.4M is safeguarded for the essential completion of the building's other functional spaces. This balanced approach ensures the Faculty of Computing becomes a center of excellence without financial overreach.

2.3 Operational Feasibility

2.3.1 Staffing & Skill Requirements

Gap Identified: Current university IT may not have specialized skills for advanced network management (Cisco DNA/SD-Access) and hybrid classroom support.

Solution:

Hire two (2) new FTE roles: a Network Specialist and an Educational Technology Specialist. Provide specialized training for existing staff on the new systems.

2.3.2 Operational Processes & Sustainability

Support Model: A clear tiered support model will be established, with Level 1 helpdesk and Level 2/3 specialist support for the faculty.

Maintenance: Scheduled maintenance windows will be established for network and server updates. A 5-year technology refresh plan will be formalized to budget for future upgrades.

User Training: Mandatory training for faculty and staff on using the hybrid classroom and lab equipment will be developed and delivered.

2.3.3 Operational Feasibility Conclusion

The project is operationally feasible. While new staffing and training are required, the organizational structure, support processes, and long-term sustainability plans are clear and achievable. The university has the capacity to operate and maintain the facility effectively.

2.4 Feasibility Overall Conclusion

The project to develop the new Faculty of Computing building is feasible from technical, economic, and operational perspectives.

Technically, it relies on proven, integratable technologies.

Economically, the investment promises a strong return through increased enrollment and enhanced reputation.

Operationally, the required support structure and processes can be established within the university's framework.

3.0 Meeting Minutes

3.1 Meeting Minutes #1

DATE/TIME		19 November 2025 20:00	
LOCATION		Online(Google Meet)	
AGENDA		1. Understand Task 2 requirements 2. Brainstorm possible questions 3. Select relevant questions 4. Task distribution	
Meeting MC		Tey Xin Yi	
ATTENDANCE			
NAME		TIME	REASON FOR ABSENCE
Tey Xin Yi		20:00	
Lee Hui Zhen		20:00	
Tan Hui Shan		20:00	
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Task 2 requirements	• Tey explained the requirements of Task 2 and the rubic. • All members reviewed the task instruction together.	Tey (19/11)
2	Question Brainstorming	• Each member proposed several potential questions. • Members discussed which questions were suitable based on the case study.	All members (19/11)
3	Question Selection	• The team discussed and shortlisted the most relevant questions. • Members agreed to refine the list before final submission.	All members (19/11)
4	Task Distribution	• Task distribution assigned as below: 1. Lee-prepare 6 questions with researched answers 2. Tey-prepare 2 questions + feasibility study 3. Tan-prepare 2 questions + meeting minutes	All members (19/11)

5	Meeting ended	10:00pm	
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3.2 Meeting Minutes #2

DATE/TIME		22 November 2025 10:00	
LOCATION		Google meet (Online)	
AGENDA		1. Review drafted answers 2. Finalize question list 3. Discuss feasibility assessment	
Meeting MC		Tey Xin Yi	
ATTENDANCE			
NAME		TIME	REASON FOR ABSENCE
Tey Xin Yi		10:00	
Lee Hui Zhen		10:00	
Tan Hui Shan		10:00	
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Explanation of drafted answers	• Lee presented the 7 answers she prepared. • Tey reviewed her 3 answers and the feasibility draft. • Tan explained the 3 answers she worked on.	All members (22/11)
2	Answer and question review	• The team reviewed all answers for accuracy, clarity, and source credibility. • Lee suggested removing questions unrelated to network communication scope. • Final selection confirmed: 10 main questions.	All members (22/11)
3	Feasibility assessment discussion	• Tey presented the feasibility study draft. • All members agreed it covered technical, economic, and operational considerations. • Minor wording improvements were suggested by the group.	Tey (22/11)
4	Meeting ended	12:00 pm	

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